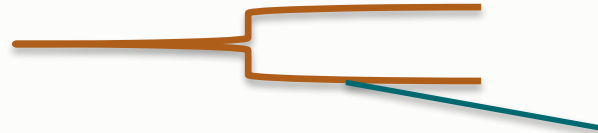


Cluster-robust variance centers uncertainty at the group level

$$\hat{V}_{\text{CR}} = (X^\top X)^{-1} \left(\sum_g X_g^\top \hat{u}_g \hat{u}_g^\top X_g \right) (X^\top X)^{-1}$$



Middle term aggregates score contributions by center.

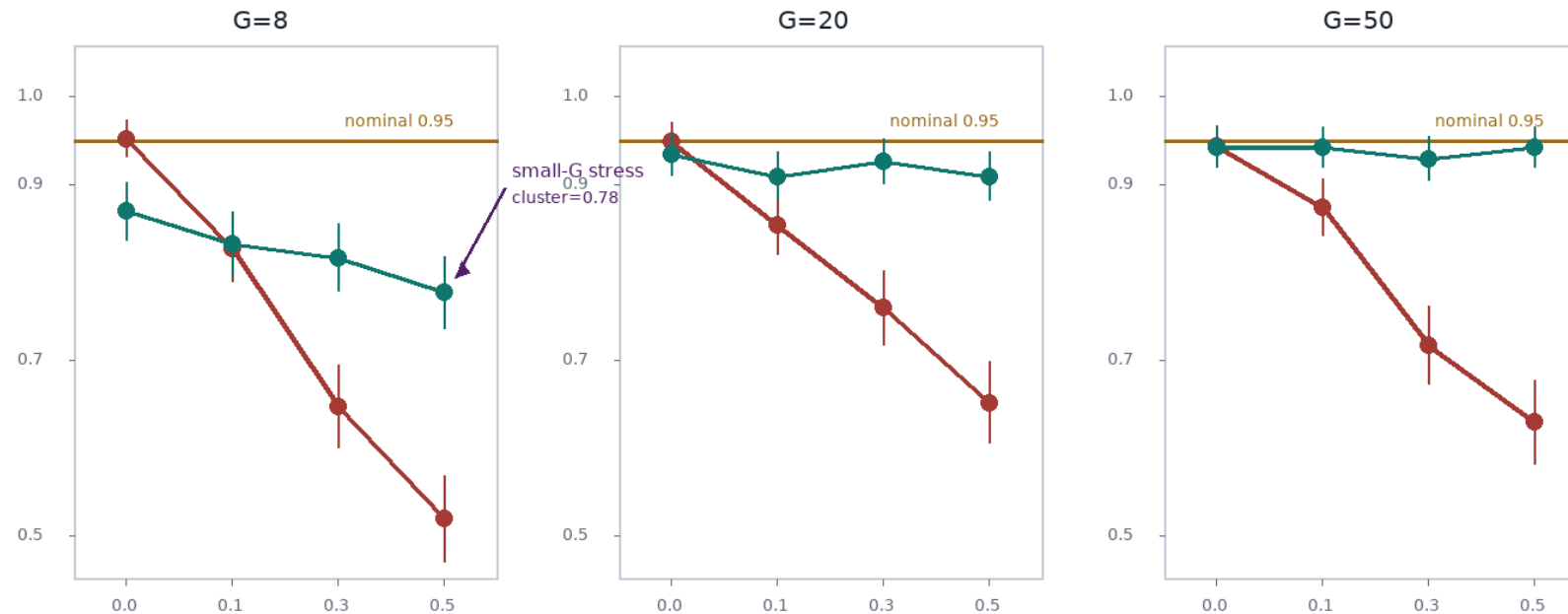
Simulation varies clustering before interval evaluation



DGP knobs feed generated samples and interval procedures.

Cluster-robust intervals recover coverage, but small-G remains stressed

Imbalanced clusters: 95% interval coverage by ICC

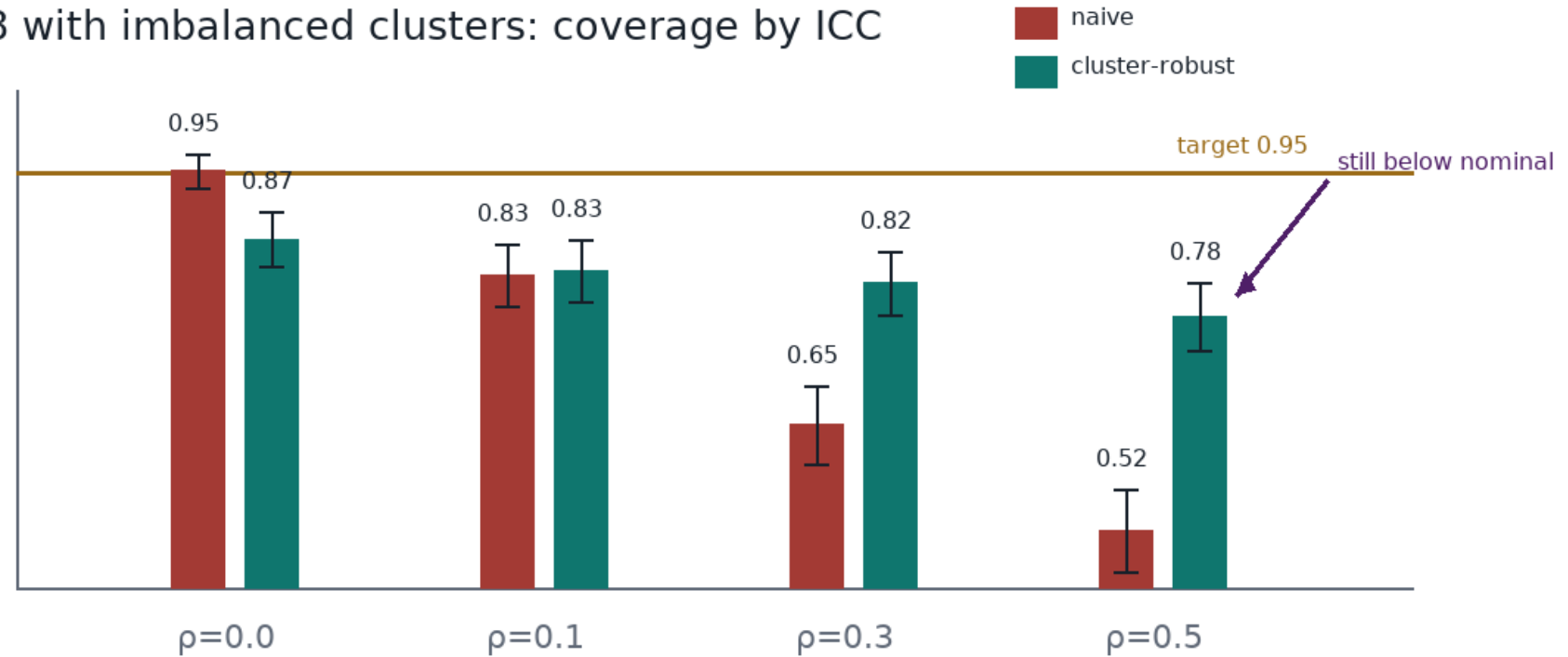


Coverage improves after center-level aggregation, with visible stress at low center counts.

Deterministic simulation; nominal 0.95 reference line shown in the plotted evidence.

Small center counts remain anti-conservative

G=8 with imbalanced clusters: coverage by ICC



Error bars show Monte Carlo ± 1.96 SE

The failure region is the small-G corner; the next experiment should target finite-sample correction.

Negative result motivates CR2 or wild cluster bootstrap as the next discriminating experiment.